

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Mechanical Engineering) SEMESTER - 3 Summer 2025 (Supply.)

Course :Bachelor of Technology (Mechanical Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 3

Subject Code & Name: BTMC302 - FLUID MECHANICS

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

12

- 1 For an ideal fluid, which of the following properties is assumed to be zero?A) DensityB) ViscosityC) PressureD) Temperature
- The upward force exerted by a fluid on a submerged object is called:A) Buoyant forceB) DragC) LiftD) Hydrostatic force
- If the metacentric height (GM) is zero, the floating body is in:A) Stable equilibriumB) Unstable equilibriumC) Neutral equilibriumD) Oscillatory motion
- Which of the following represents the motion of fluid particles without considering the forces causing the motion?A) Fluid dynamicsB) Fluid staticsC) Fluid kinematicsD) Hydrostatics
- 5 In the Lagrangian method, fluid motion is studied by,A) Observing fixed points in space B) Tracking individual fluid particlesC) Measuring fluid velocity at specific locationsD) Analyzing streamlines
- 6 Which of the following forces are considered in Euler's equation for inviscid flow?A) Viscous and pressure forcesB) Pressure, gravity, and surface tensionC) Inertial, pressure, and viscous forcesD) Gravity and pressure forces
- The total head in Bernoulli's equation is the sum of:A) Static and dynamic pressureB) Pressure head, velocity head, and elevation headC) Pressure energy and internal energyD) Kinetic energy and potential energy only
- The velocity profile of laminar flow in a circular pipe is:A) ParabolicB) FlatC) LinearD) Uniform
- For a Newtonian fluid in fully developed laminar flow inside a pipe, the shear stress is maximum:A) At the pipe wallB) At the pipe centerC) Uniform across the pipeD) Just near the pipe center
- 10 Which shape typically produces the highest lift-to-drag ratio?A) Flat plateB) SphereC) Streamlined airfoilD) Cube
- 11 The number of π terms formed using Buckingham's π -theorem is:A) Equal to the number of variablesB) Equal to number of repeating variablesC) Equal to number of fundamental dimensionsD) Equal to number of variables minus number of fundamental dimensions

- 12 The physical significance of the Froude number is: A) Ratio of inertia to viscous forces B) Ratio of pressure to gravitational forces C) Ratio of inertia to gravity forces D) Ratio of inertia to pressure forces

Q2. Solve the following.

- A) Define the Viscosity and derive the Newton's law of viscosity. 6
 B) In a stream of glycerin in motion at a certain point the velocity gradient is 0.25 s^{-1} . If for fluid density, $\rho = 1268.4 \text{ kg/m}^3$ and kinematic viscosity $\nu = 6.30 \times 10^{-4} \text{ m}^2/\text{s}$, calculate the shear stress at the point. 6

Q3. Solve the following.

Explain in detail any two of the following: Uniform and non uniform flow, rotational and irrotational flow, velocity potential and flow net. 6

A horizontal venturimeter with inlet and throat diameters 30 cm and 15 cm respectively is used to measure the flow of water. The reading of differential manometer connected to the inlet and the throat is 10 cm mercury. Determine the rate of flow. Take $C_d = 0.98$. 6

Q4. Solve Any Two of the following.

- A) Explain the major losses in case of flow through pipes and also write the Darcy's and Chezy's equations to calculate energy losses due to friction with clearly mention of each term in the equation. 6

A crude oil of viscosity 0.9 poise and specific gravity 0.8 is flowing through a horizontal circular pipe of diameter 50 mm and length 15 m. Calculate the difference of pressure at the two ends of the pipe, if 50 kg of the oil is collected in a tank in 15 seconds. 6

Write a short note on Hydraulic Gradient Line (HGL) and Total Energy Line (TEL). 6

Q5. Solve Any Two of the following.

- A) Explain the concept of the boundary layer in fluid flow. Discuss the development of the boundary layer over a flat plate with sketch. 6

- B) Define the terms lift, drag, lift coefficient and drag coefficient. 6

- C) Explain the various components of drag: pressure drag, skin friction drag, induced drag. Give examples of where each dominates. 6

Q6. Solve Any Two of the following.

State and explain Buckingham's π -theorem. Outline the steps involved in applying it to reduce a physical problem into dimensionless form. 6

Define Reynolds number. Derive its expression using dimensional analysis and explain its role in determining the nature of fluid flow. 6

8. Find the velocity of flow of an oil through a pipe, when a difference of mercury level in differential manometer connected to two tapings of pitot tube is 15 cm. Take sp. gr. of oil as 0.8 and coe. of pitot tube as 0.98 6

*** End ***